



Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

Participant Name: Umer Khan (Leader, Team: Quantum Coders)

Problem Statement: [AI for Better Living and Smarter Communities]

Develop an AI-powered Unified Emergency Response & Disaster Management Platform for India that enables intelligent emergency reporting, AI-based incident verification, real-time coordination, and preventive hazard detection to build safer and smarter communities.

NSERS – National Smart Emergency Response System

NSERS (National Smart Emergency Response System) is an AI-powered unified emergency response and disaster management platform designed to make communities safer and emergency services smarter. Citizens can report accidents, disasters, crimes, or public hazards using **text, voice, images, or videos**, while the system automatically captures the incident location through GPS and Google Maps.

Using **Google Gemini**, the platform analyzes and verifies reports, identifies the type and severity of the emergency, filters potential false alerts, and recommends the appropriate response agencies such as **Police, Ambulance, Fire Brigade, NDRF, and Disaster Management Authorities**. It also provides authorities with a centralized real-time dashboard for faster coordination, resource allocation, and live incident tracking.

Beyond emergency response, NSERS enables **preventive hazard reporting**, allowing citizens to report risks before they escalate into disasters. By combining **AI, location intelligence, and real-time collaboration**, NSERS aims to reduce response time, improve public safety, and build **safer, smarter, and more resilient communities across India**.

1. Our Approach & Google Cloud Technologies:

We developed **NSERS (National Smart Emergency Response System)** using **Google Gemini** for multimodal AI analysis of text, images, videos, and voice reports, **Google Maps Platform** for real-time location intelligence and nearest resource identification, **Vertex AI** for AI model deployment, **Firebase** for authentication, notifications, and real-time updates, **Cloud Storage** for secure media storage, and **BigQuery** for analytics and disaster trend insights. The platform leverages AI to verify incidents, classify emergencies, recommend response agencies, and streamline emergency coordination.

2. Real-World Problem & Impact:

India's emergency response system is fragmented, requiring citizens to contact multiple helplines while authorities often lack real-time situational awareness. NSERS provides a **single AI-powered platform** that reduces response time, improves inter-agency coordination, minimizes false reports, enables preventive hazard reporting, and enhances public safety. It benefits **citizens, emergency responders, disaster management authorities, hospitals, and government agencies**, contributing to safer and smarter communities.

3. Core Architecture & Intelligent Workflow:

Citizen Report (Text/Image/Video/Voice) → GPS & Google Maps → Gemini AI Verification → Incident Classification & Severity Assessment → AI Confidence Score → Dispatch Recommendation (Police/Ambulance/Fire/NDRF) → Authority Dashboard → Resource Allocation → Live Response Tracking → Incident Resolution → BigQuery Analytics & Future Disaster Prediction

Opportunities

How is NSERS different from existing solutions?

1. **Unified emergency platform** replacing fragmented emergency reporting systems.
2. **AI-powered multimodal verification** using text, voice, images, and videos to reduce false alerts.
3. **Automatic GPS & Google Maps integration** for precise incident location and faster response.
4. **Intelligent dispatch recommendations** that suggest the nearest Police, Ambulance, Fire Brigade, or NDRF based on incident type and severity.
5. **Preventive hazard reporting** empowers citizens to report risks before they become disasters.
6. **Real-time coordination and live tracking** between citizens and emergency authorities.

Unique Selling Proposition (USP)

1. **Gemini AI** for multimodal incident understanding and verification.
2. **Single platform** for reporting all emergencies across India.
3. **AI-driven smart dispatch** with location intelligence and route optimization.
4. **Live response tracking** for complete transparency.
5. **Fraud detection & AI Confidence Score** to prioritize genuine emergencies.
6. **BigQuery-powered analytics** for hotspot identification and predictive disaster preparedness.

Features

Emergency Response	AI Intelligence	Smart Operations	Analytics & Prevention
One-Tap Emergency SOS	Gemini AI Multimodal Analysis	Authority Command Dashboard	BigQuery Analytics
Emergency Reporting (Text, Voice, Image & Video)	AI Incident Classification	Intelligent Dispatch Recommendation	Disaster Hotspot Detection
Automatic GPS Location Detection	Severity Assessment	Resource Allocation	Predictive Disaster Insights
Google Maps Integration	AI Confidence Score	Live Incident Tracking	Preventive Hazard Reporting
Emergency Contact Alerts	Fraud & Fake Report Detection	Multi-Agency Coordination	Response Time Analytics
Nearby Hospitals & Emergency Services	AI Incident Summary	Role-Based Access Control	Department Performance Reports
Manual Location Pinning	Duplicate Report Detection	Route Optimization	Risk Assessment Dashboard
Incident Status Updates	Voice-to-Text Processing	Unit Availability Monitoring	Community Safety Insights

Process flow diagram



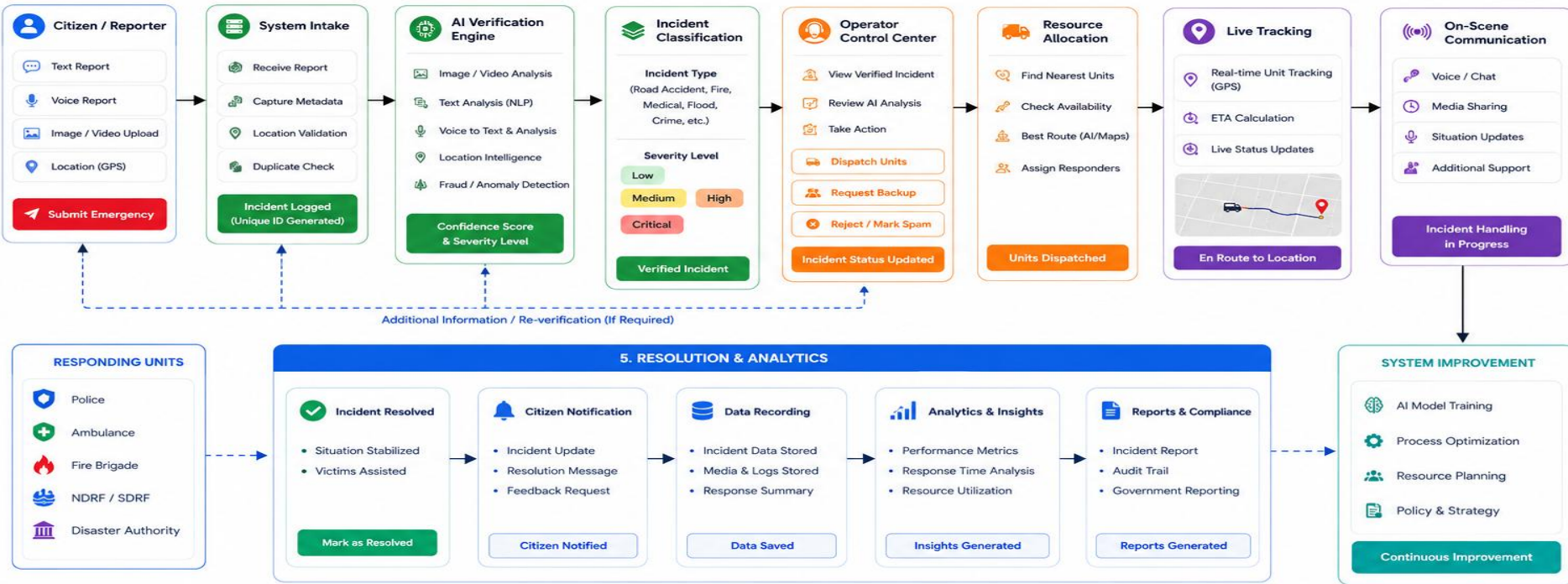
1. INCIDENT INITIATION

2. AI VERIFICATION & ANALYSIS

3. DISPATCH & RESPONSE

4. LIVE TRACKING & COMMUNICATION

5. RESOLUTION & ANALYTICS



KEY BENEFITS



Faster Response

AI-powered verification & smart dispatch



Accurate Decision

Multimodal AI analysis & confidence scoring



Resource Optimization

Right unit, right time, right location



Real-time Tracking

Live updates & better coordination



Transparency

Audit trail, reports & accountability



Data-driven Insights

Analytics for improved preparedness

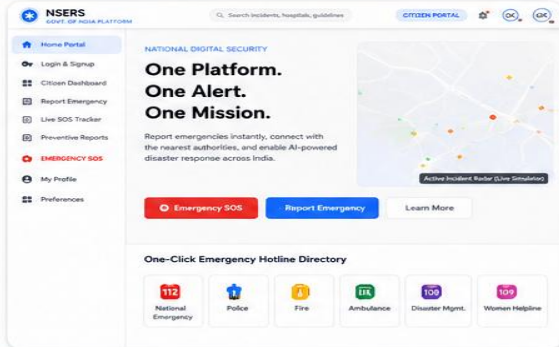
Wireframe

PART 1

1. Home Portal (Landing Page)

Unified entry point for citizens with alerts, quick actions and emergency hotlines.

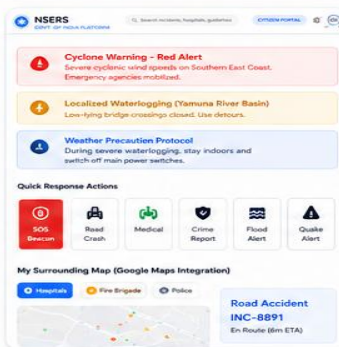
- Real-time disaster alerts
- Quick emergency actions
- Google Maps live radar
- One-click emergency hotlines



PART 2

2. Citizen Dashboard

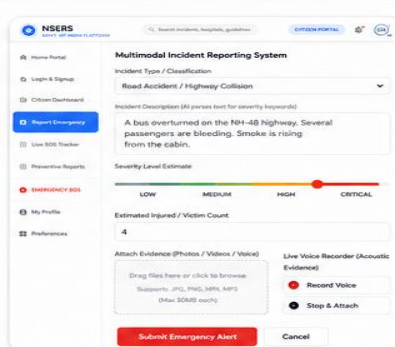
Live feed of alerts, nearby resources, quick actions and incident status.



PART 3

3. Report Emergency

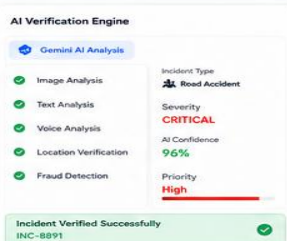
Multimodal incident submission with AI severity estimation.



PART 4

4. AI Verification & Classification

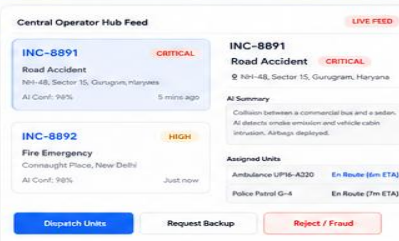
AI engine analyzes and verifies the report, assigning confidence and severity.



PART 5

5. Operator Control Center

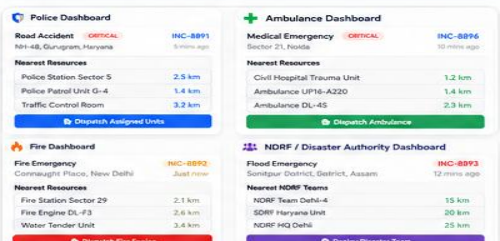
Operators review, take action and dispatch the right units.



PART 6

6. Department Dashboards (Unit Dispatch)

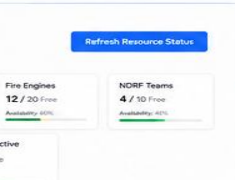
Each department receives tasks and manages unit response.



PART 7

7. Resource Management Portal

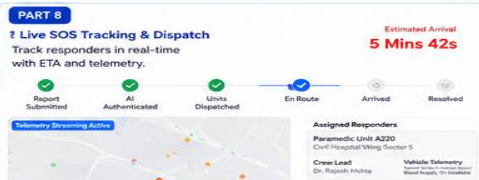
Real-time overview of national resources and availability.



PART 8

8. Live SOS Tracking & Dispatch

Track responders in real-time with ETA and telemetry.



PART 9

9. Incident Resolved

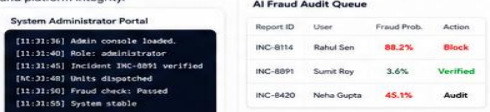
Incident closure, summary & public awareness.



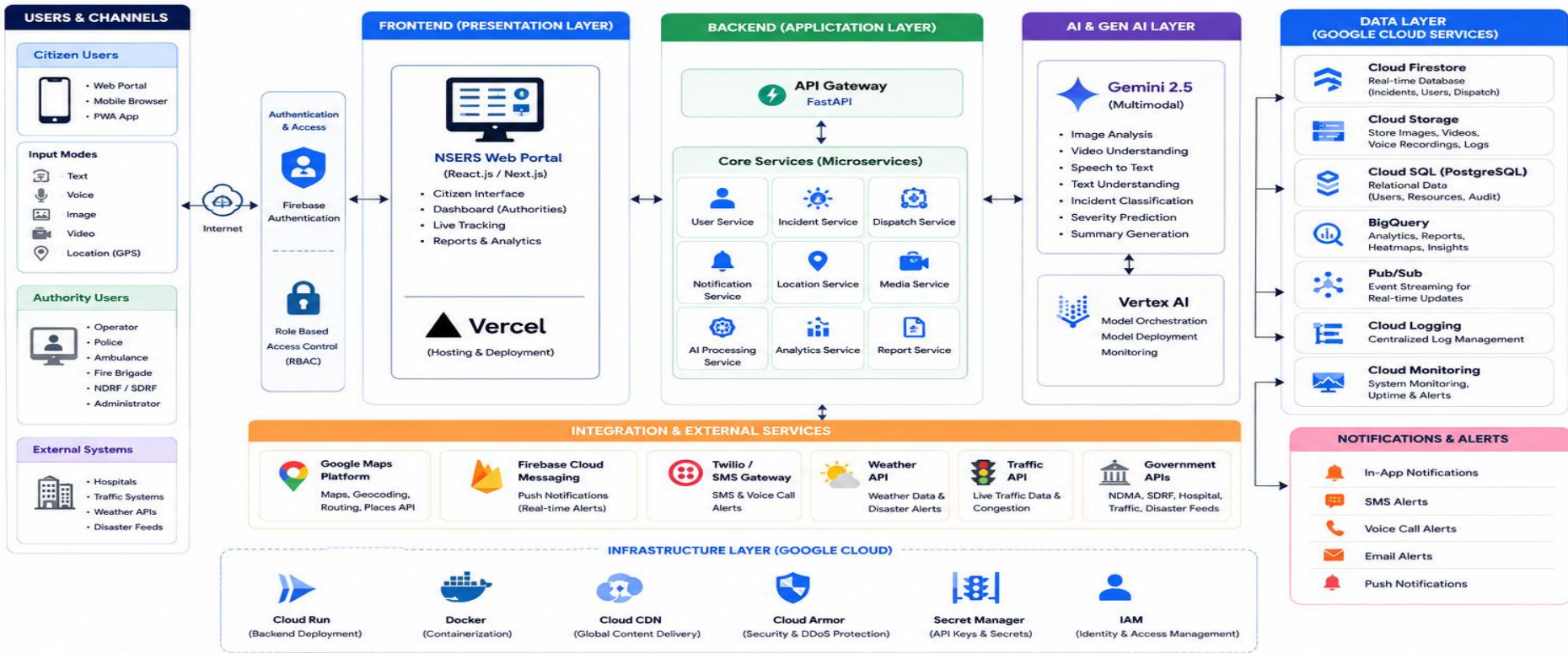
PART 10

10. Administrator Portal

System monitoring, fraud audit and platform integrity.



Architecture diagram



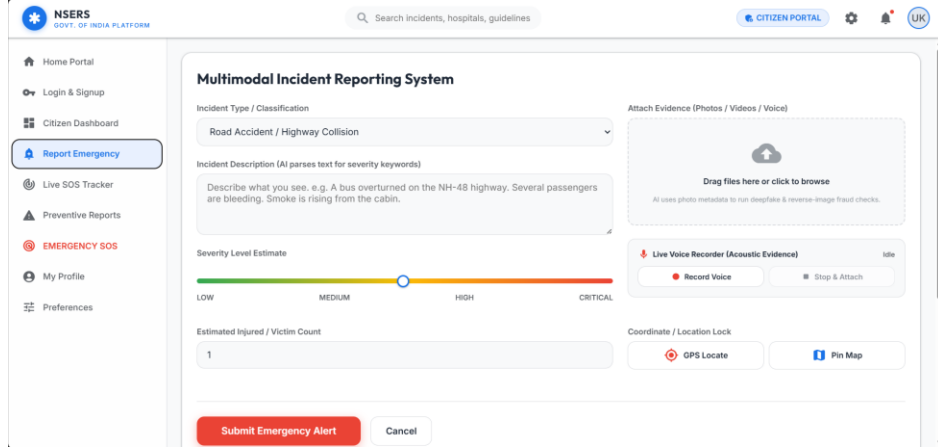
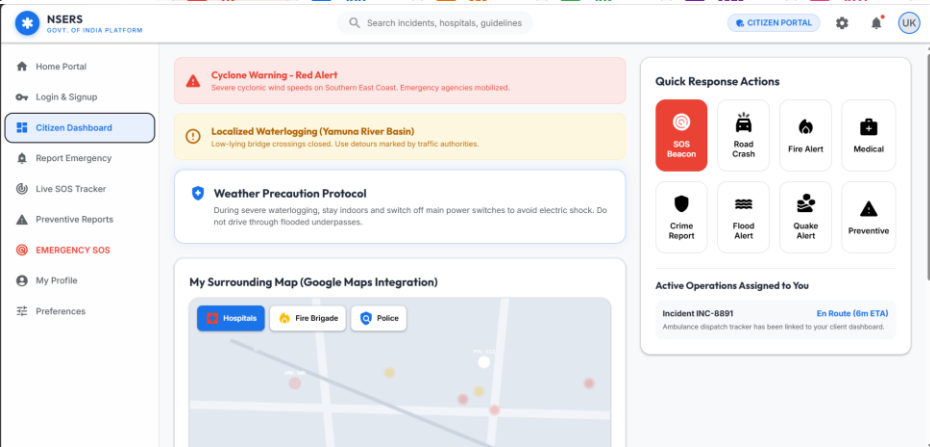
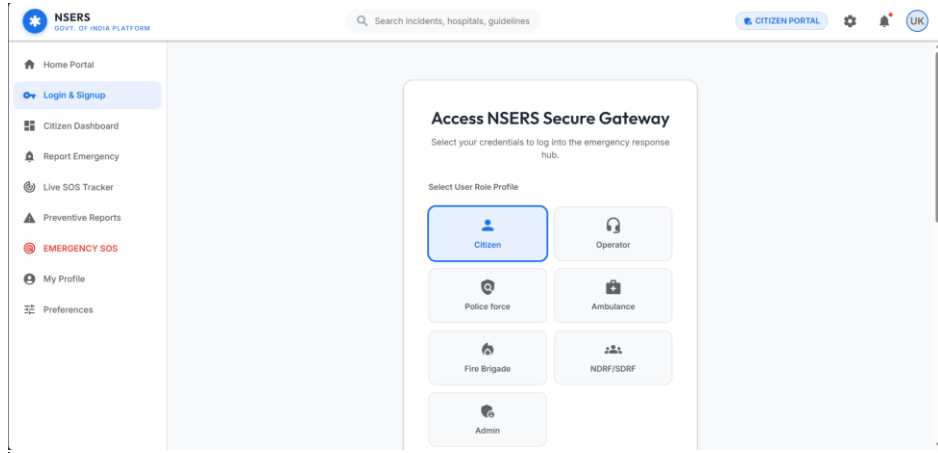
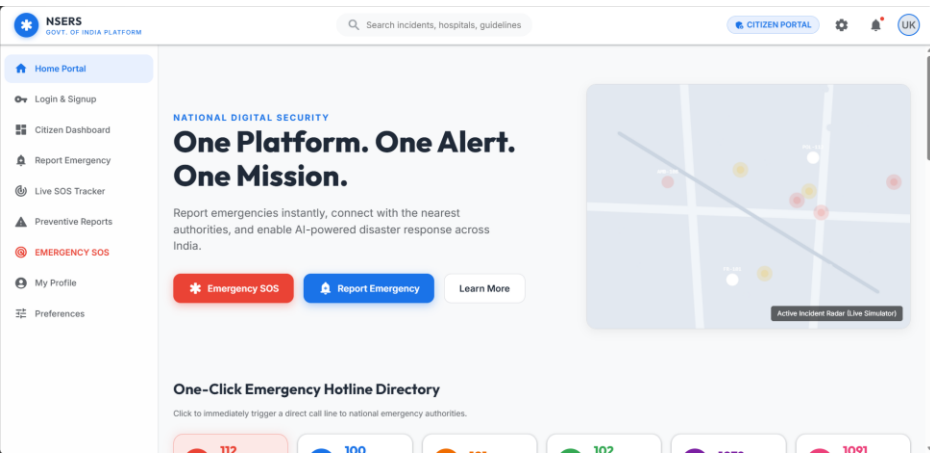
FLOW SUMMARY



Technology / Service	Purpose & Justification
Google Gemini 2.5	Multimodal AI for analyzing text, images, videos, and voice to classify incidents, verify reports, generate AI summaries, and recommend emergency response actions.
Vertex AI	Deploys, manages, and scales AI models, enabling reliable inference and future model improvements without changing the application architecture.
Google Maps Platform	Provides GPS location, reverse geocoding, interactive maps, nearest emergency unit identification, route optimization, and ETA calculation.
Firebase Authentication	Secure authentication for citizens and emergency personnel using Google Sign-In, email, or phone-based login with role-based access control.
Cloud Firestore	Stores real-time incident reports, dispatch status, user information, and live updates with instant synchronization across all dashboards.
Cloud Storage	Securely stores uploaded images, videos, voice recordings, and evidence while supporting scalable multimedia management.
Cloud Run	Hosts backend APIs and AI services with automatic scaling, ensuring reliable performance during high emergency traffic.
BigQuery	Performs large-scale analytics to identify disaster hotspots, response trends, department performance, and predictive risk insights.
Cloud Monitoring & Logging	Continuously monitors system health, API performance, error logs, and operational metrics for a reliable production environment.
Cloud Pub/Sub	Enables real-time event-driven communication between reporting, AI processing, dispatch, notifications, and authority dashboards.

Technologies Used

Category	Technology	Purpose
Frontend	React.js, Next.js, TypeScript, HTML5, CSS3, Tailwind CSS	Responsive citizen portal & authority dashboard
Backend	Python (FastAPI)	REST APIs, business logic & AI orchestration
AI & ML	Python, Google Gemini API	Multimodal incident analysis, classification & AI-powered decision making
Database	PostgreSQL, Cloud Firestore	Store users, incidents, emergency units & real-time data
Real-Time Communication	WebSockets, Firebase Cloud Messaging (FCM)	Live tracking, notifications & instant updates
Maps & Location	Google Maps Platform	GPS, routing, geocoding & nearest emergency unit identification
Deployment & DevOps	Docker, GitHub Actions	Containerized deployment & CI/CD automation



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CITIZEN PORTAL

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UK

Home Portal

Login & Signup

Citizen Dashboard

Report Emergency

Live SOS Tracker

Preventive Reports

EMERGENCY SOS

My Profile

Preferences

Live Dispatch Telemetry

Incident ID: INC-8891 (Ambulance UP16-A220 Assigned)

Estimated Arrival
5 Mins 42s

Report Submitted

AI Authenticated

Units Dispatched

En Route

Arrived

Resolved

Telemetry Streaming Active

Assigned Responders

Paramedic Unit A220

Civil Hospital Wing Sector 5

Crew Lead: Dr. Rajesh Mehta

Vehicle Telemetry: Speed: 64 km/h (Active Siren)

Emergency Medical Care: Blood Supply 0+ Available

Call Unit Lead

Details

Trauma Broadcast Active

Nearby traffic is receiving automated alerts via Android Emergency Location services to yield priority route access.

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CITIZEN PORTAL

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UK

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EMERGENCY SOS

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Hazard & Disaster Prevention Log

Submit hazard vulnerabilities (e.g. open electrical wiring, damaged bridges) to prevent safety issues before they escalate.

Hazard Classification Type

Broken Bridge Railing / Structural Damage

Vulnerability Location Details

e.g. Outer Ring Road, Near Flyover pillar 88

Infrastructure Severity Level

Critical Vulnerability (High risk of imminent danger)

Comments / Additional Safety Hazards

Provide any additional comments...

Submit Preventive Hazard Log

Active Reported Infrastructure Concerns

Open High Tension Wire

Location: Noida Sector 62. Reported 3 days ago.

Status: Assigned to DISCOM Wing

Yamuna Canal Railing

Location: Yamuna Expressway Bridge. Reported 1 day ago.

Status: Inspection Scheduled

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ADMINISTRATOR

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Home Portal

Login & Signup

Admin Console

My Profile

Preferences

System Administrator Portal

Review system integrity logs, user credentials verification, and fraud detection levels.

Real-time System Logs

```
[1:31:36 am] - Admin console view loaded.
[1:31:36 am] - Navigated to admin-panel
[1:31:36 am] - Active role switched to administrator
[1:31:38 am] - Navigated to login-signup-page
[1:31:38 am] - Navigated to emergency-sos-page
[1:30:38 am] - Navigated to hazard-report-page
[1:29:35 am] - Navigated to live-tracking-page
[1:28:29 am] - Navigated to report-emergency-page
[1:27:48 am] - Navigated to citizen-dashboard
[1:25:59 am] - Navigated to login-signup-page
[1:25:26 am] - NSERS Core services started. 15 Screens loaded.
23:01:05 - System initialized in New Delhi HQ.
```

AI Fraud Flag Audit Queue

Report ID	Citizen User	Fraud Probability	Action Link
INC-8114	Rahul Sen	88.2% (Spam)	Block
INC-8891	Sumit Roy	3.6% (Verified)	N/A
INC-8420	Neha Gupta	45.1% (Suspected)	Audit

NSERS HUB

A joint safety initiative by the Ministry of Electronics and Information Technology (MeitY) and the National Disaster Management Authority (NDMA) in technical partnership with Google India.

Quick Directories

National Portal Home

Relief Camp Locator

NDMA Safety Guidelines

Emergency Helplines

National Emergency: 112

Disaster Management: 1078

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OPERATOR

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UK

Home Portal

Login & Signup

Control Dashboard

Resources (NDRF)

National Analytics

My Profile

Preferences

National Resource Allocation Portal

Review and manage active ambulances, police forces, and NDRF teams.

Refresh Resource Status

Ambulance Units

18 / 50 Free

Availability: 36%

Capacity in use: 32

Police Patrol Vehicles

42 / 100 Free

Availability: 42%

Capacity in use: 58

Fire Engines

12 / 20 Free

Availability: 60%

Capacity in use: 8

NDRF Squad Teams

4 / 10 Free

Availability: 40%

Capacity in use: 6

Hospital ICU Beds Available

142 / 500 Free

Availability: 35%

Capacity in use: 358

Paramedics Active

60 / 200 Free

Availability: 30%

Capacity in use: 140

Regional Resource Allocation Map

Zone North (Delhi NCR)

Available units: 82% capacity. No active disaster strain.

Zone South (Chennai Wing)

Available units: 35% capacity. Heavy strain due to flood monitoring.

Zone East (Assam Wing)

Available units: 62% capacity. NDRF battalions fully deployed.

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OPERATOR

6.2 Min
AVERAGE RESPONSE TIME

6 Active Logs
PENDING DISPATCHES

42 Dispatchers
RESPONDERS AVAILABLE

18 Teams busy
RESPONDERS BUSY

Home Portal

Login & Signup

Control Dashboard

Resources (NDRF)

National Analytics

My Profile

Preferences

Central Operator Hub Feed LIVE FEED

INC-8891 CRITICAL
Road Accident
NH-48, Sector 15, Gurugram, Haryana
AI Conf: 98%
5 mins ago

INC-8892 HIGH
Fire Emergency
Commercial Complex, Block C, Connaught Place, New Delhi
AI Conf: 94%
Just now

INC-8893 CRITICAL
Flood Emergency
Tejpur Outer Suburbs, Sonitpur District, Assam
AI Conf: 98%
12 mins ago

Road Accident CRITICAL INC-8891
5 mins ago
NH-48, Sector 15, Gurugram, Haryana

Incident Evidence Files
Evidence

Nearest Emergency Resources
Civil Hospital Trauma Unit 1.2 km away
Police Station Sector 5 2.5 km away
Fire Headquarters CP 4.1 km away

Operational Controls
Dispatch Assigned Units
Reject Fraud Alert
Request Backup

AI Summary & Intelligence
Collision between a commercial bus and a sedan. AI detects smoke emission and vehicle cabin intrusion. Airbags deployed.

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POLICE

5.1 Min
AVERAGE RESPONSE TIME

2 Accidents
PENDING DISPATCHES

180 Officers
RESPONDERS AVAILABLE

42 Units
RESPONDERS BUSY

Home Portal

Login & Signup

Control Dashboard

Resources (NDRF)

National Analytics

My Profile

Preferences

Police Dispatch Feed LIVE FEED

INC-8891 CRITICAL
Road Accident
NH-48, Sector 15, Gurugram, Haryana
AI Conf: 98%
5 mins ago

INC-8895 HIGH
Crime / Security
Gold Shop jewelry market, Karol Bagh, Delhi
AI Conf: 97%
3 mins ago

Road Accident CRITICAL INC-8891
5 mins ago
NH-48, Sector 15, Gurugram, Haryana

Incident Evidence Files
Evidence

Nearest Emergency Resources
Civil Hospital Trauma Unit 1.2 km away
Police Station Sector 5 2.5 km away
Fire Headquarters CP 4.1 km away

Operational Controls
Dispatch Assigned Units
Reject Fraud Alert
Request Backup

AI Summary & Intelligence
Collision between a commercial bus and a sedan. AI detects smoke emission and vehicle cabin intrusion. Airbags deployed.

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AMBULANCE

7.4 Min
AVERAGE RESPONSE TIME

2 Medical Cases
PENDING DISPATCHES

18 Free ICU Beds
RESPONDERS AVAILABLE

32 Ambulances
RESPONDERS BUSY

Home Portal

Login & Signup

Control Dashboard

Resources (NDRF)

National Analytics

My Profile

Preferences

Medical Trauma Dispatch Feed LIVE FEED

INC-8891 CRITICAL
Road Accident
NH-48, Sector 15, Gurugram, Haryana
AI Conf: 98%
5 mins ago

INC-8896 CRITICAL
Medical Emergency
Residential Complex, Sector 21, Noida
AI Conf: 95%
10 mins ago

Road Accident CRITICAL INC-8891
5 mins ago
NH-48, Sector 15, Gurugram, Haryana

Incident Evidence Files
Evidence

Nearest Emergency Resources
Civil Hospital Trauma Unit 1.2 km away
Police Station Sector 5 2.5 km away
Fire Headquarters CP 4.1 km away

Operational Controls
Dispatch Assigned Units
Reject Fraud Alert
Request Backup

AI Summary & Intelligence
Collision between a commercial bus and a sedan. AI detects smoke emission and vehicle cabin intrusion. Airbags deployed.

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DISASTER AUTHORITY

RIVER WATER LEVEL ADVISORY
Tejpur Outer Suburbs river water levels exceeded danger line by 12 meters. Alert coastal NDRF battalions.

18.2 Min
AVERAGE RESPONSE TIME

1 Flood Outbreak
PENDING DISPATCHES

4 Battalions Free
RESPONDERS AVAILABLE

6 NDRF Teams
RESPONDERS BUSY

Home Portal

Login & Signup

Control Dashboard

Resources (NDRF)

National Analytics

My Profile

Preferences

NDRF Disaster Relief Feed LIVE FEED

INC-8893 CRITICAL
Flood Emergency
Tejpur Outer Suburbs, Sonitpur District, Assam
AI Conf: 98%
12 mins ago

Flood Emergency CRITICAL INC-8893
12 mins ago
Tejpur Outer Suburbs, Sonitpur District, Assam

Incident Evidence Files
Evidence

Nearest Emergency Resources
Civil Hospital Trauma Unit 1.2 km away
Police Station Sector 5 2.5 km away
Fire Headquarters CP 4.1 km away

Operational Controls
Dispatch Assigned Units

AI Summary & Intelligence

Google Cloud 

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Thank you

